

Lee Hart's "H8 Mini" Miniature Clone of the Heathkit H8 Computer

Instructions for the Enhanced Keycaps

PRS 8-6-26

Overview

The "H8 Mini" retro-computer kit comes with 16x small tactile switches, which comprise the keypad array. These are tiny tactile pushbutton switches, too small to have key legends or labels, so that information is on the PCB in the form of silkscreened text.....this can be rather hard to read when actually trying to use the computer via its keypad. In addition, there is not enough space on the PCB to include ALL the information on secondary key functions (the original H8 key labels had as much as three lines of text per key). Furthermore, pressing on those tiny round switch actuators for all the time it can take to enter even a small program can be uncomfortable.

Accordingly, a parallel project was conceived to offer a set of keypad enhancements to give a better user experience with the H8 Mini. This project includes the following:

- A different version of the original kit's tactile switch, having a taller actuator
- 3D printable 'keycaps' which will press down over the taller switch actuators, giving each key a larger, more comfortable 'face' to press on, and that surface will have enough room to support a label that is large enough to include all of the primary and secondary text
- Laser-printable labels to fit the printed keycap 'faces', each label including the original (or slightly altered) text from the original H8 keypad labels
- Not directly related to the H8 Mini keypad is the associated LED display filter, which will give the display a much clearer appearance

At its basic level, these instructions, along with the STL file for the 3D printed keycaps and the PDF file for printing the keycap labels, will be shared with all buyers of the H8 Mini kit. At a higher level, a complete kit (probably facilitated through eBay) will be made available, consisting of 16x taller tactile switches, 16x 3D printed keycaps, one sheet of printed keycap labels, and a pre-cut piece of red filter material for the display.

Tall Tactile Switches

The 3D printed keycaps require tactile switches with actuators that are taller than the short actuators on the 16 switches furnished as part of the "H8 Mini" kit. These are alternate switches are very inexpensive and readily available.

Whereas the original switches are TE Connectivity/Alco Switch # 1825910-6 (the -6 designates a switch requiring a press-force of 1.6 Newtons in order to actuate), which Digikey sells as 450-1650-ND. The alternate taller switches can be either of the two following models:

- TE Connectivity/Alco Switch # 1-1825910-4, Digikey 450-1653-ND (1.6 N required)
- TE Connectivity/Alco Switch # 1-1825910-5, Digikey 450-1940-ND (2.6 N required)

Note the lack of a leading “1-” on the original part number, and presence of the leading “1-” on the two alternate part numbers; this is a critical difference which must be observed.

These have the same footprint and pinout as the originally supplied switches, but have actuators that are 4.5mm taller. Digikey typically has 10’s of thousands of the first number in stock, and hundreds of the second number. These can be differentiated from one another by the first switch’s black actuator and the second switch’s red actuator.

A total of 16 of these taller switches are required to populate the keypad area of the PCB.

Make absolutely sure that these switches have been pressed down firmly into their four pin holes on the PCB, and are thus flat against the front surface of the PCB. Before soldering the pins, view across each row and down each column of the switch array, looking for actuators that are leaning out of alignment with the others, and correct this before soldering.

3D Printable Keycaps

The 3D printable ‘keycaps’ are designed to fit, using ‘press on’ friction, the taller tactile switches described above. The keycap design has three features”

- Flat front surface, or ‘face’ (ideal for affixing printed keycap labels) measuring 10x10mm square
- Square ‘fence’ that surrounds the top half of the body of the tactile switch, with a close tolerance fit; this serves to prevent keycap rotation on the round switch actuator
- Round hole into which the round (cylindrical) switch actuator is inserted. The actuator is actually slightly tapered, and the hole is cylindrical, sized to that when the end of the actuator goes all the way in and touches the back side of the front face, the bottom edge of the hole will wedge firmly onto the tapered surface of the actuator. By this means, the keycap should not come loose or fall off on its own, yet it can be pulled off of the switch actuator by using a small amount of force; the outer dimension of the ‘fence’ is smaller than the outer edge of the keycap face, giving an overhanging lip that is useful for grabbing the keycap when removing it.

The keycap is ‘defined’ by a single “STL” (stereolithography “Standard Triangle Language” or “Standard Tessellation Language”) file named “H8 Mini Keycap.stl”. This can be downloaded from the following ‘fileshare’ webpage under the “H8 Mini” heading.:

www.serpentwebsite.com/fileshare.htm

To produce the actual keycaps, import the STL file into the ‘slicer’ software associated with the 3D printer. Make sure to place the virtual keycap object on the slicer’s workplane in its default orientation, i.e. ‘face down’. Visually verify that the virtual keycap object is ‘wide side down, narrow end up’. It is recommended that a single keycap then be printed, and inspected, then test fitted onto one of the tall tactile switches to make sure that it ‘stays put’ and moves freely as the keycap is pressed to operate the tactile switch (it should not get stuck, or bind). It may be necessary on some printers/slicers to tweak the print settings to get a good result. Note that the keycap face needs to come out flat and smooth to the touch (for best label fit and adhesion), although it is OK and normal for the face to have visible seams where multiple passes of the printer’s nozzle took place.

Once the above test has been done successfully, copy that original imported virtual keycap object 15 times, for a total of 16 keycap objects arranged in some sort of array on the slicer’s workplane; a good method is to arrange them in a 4 x 4 array or a single row. Then command the printer to print the set of 16 keycaps.

Regular “PLA” filament is a good choice for the 3D printing, as it is readily available and does not produce noxious odors during the printing process. ABS would be better, but that material requires active ventilation around the printer.

Since the printed keycap labels are black, it is best to use black filament when printing the keycaps.

There is a tiny round hole in the center of the keycap face. This serves two purposes:

- Allows displaced air to escape easily when inserting the switch actuator into the hole on the keycap base
- If there ends up being excessive keycap wobble, a drop of glue can be wicked down through this hole into the space around the tip of the actuator. If this is done, a removable, non-gripping resin such as the BLUE Loctite would be better than something like CA (Super) glue. Make sure that no ‘glue’ makes its way down to the switch bodies.

Keycap Labels

A set of keycap labels have been designed, and may be printed as a 4x4 array of 16 labels on a single sheet of paper or equivalent film. Using actual wood-based paper is NOT recommended due to fuzziness of the printed image, general rougher appearance, and lack of durability during use of the keypad. A better choice is one of the several laser-printable synthetic ‘paper’ substitutes such as 3M “Print To Last” or Xerox “Revolution Never Tear”. Unfortunately, these products are normally sold only in boxes of 100 sheets, and such a box might cost several tens of dollars. Another similar product is Avery #60501 “UltraDuty GHS Chemical Labels”, although it has the disadvantage of having its own adhesive and backing paper which needs to be laboriously peeled off of each small keycap label before it can be applied, and the grip of the adhesive is not quite as durable as some other adhesives that might be used with the first two

products; this product also has a lightly textured/matte surface, which might or might not be desirable in the view of the kit builder (the matte surface makes the small text look a bit fuzzy).

All of the above products have the following pertinent characteristics:

- Must be printed on using a laser printer (NO inkjet)
- Toner is unlikely to come off
- Tough material
- Dimensionally stable (does not shrink, stretch, curl)
- Abrasion resistant
- Moisture resistant
- Chemical & skin oils resistant
- Very durable and excellent for keycap labels

There are three available keycap label design variants, downloadable from the same ‘fileshare’ webpage as the 3D keycap ‘STL’ file:

- H8 Mini keycap labels-MK3 larger secondary text-FINAL.pdf
 - o Black label, white primary characters, yellow secondary characters
 - o This is probably the clearest, nicest looking version
 - o Requires the use of a color laser printer
- H8 Mini keycap labels-MK3 larger secondary text-all white-FINAL.pdf
 - o Black label, white primary characters, white secondary characters
 - o Can be printed on a black & white (monochrome) laser printer
- H8 Mini keycap labels-MK3 larger secondary text-all white-grey back-FINAL.pdf
 - o Charcoal gray label, white primary characters, white secondary characters
 - o Closest appearance to the original H8’s keypad labels
 - o Requires the use of a color laser printer

These new labels have the same text, both primary and secondary, as the original H8 keypad labels, with the following exceptions:

- The label for the key having the primary text “7” has secondary text of “STEP” instead of “SI”, which meant Step-Increment or something similar. The meaning is the same.
- The label for the key having the primary text “#” has secondary text of “MEM+0” instead of “RTM / 0”. The meaning is the same.
- The label for the key having the primary text “/” has secondary text of “RESET+0” instead of “RST / 0”. The meaning is the same.

In order to apply the labels to the keycaps, first place the keycaps on the tactile switch actuators in all 16 positions of the keypad. Then print the label pattern from the selected PDF file, making sure to select “Full Size” or similar in the driver for the laser printer. The array of 16 labels will be printed in a pattern of 4 x 4 labels on a single sheet of the laser-printable material, as described above. Notice that a faint outline surrounds the label pattern; use a scissors to cut out the pattern along those lines, leaving a 2mm-wide white margin around the pattern.

This results in the label pattern with a white grid of 2mm-wide lines separating the 16 labels. These lines serve a purpose; they are easier to see clearly when cutting the individual labels from the overall pattern, AND the person using the scissors can more precisely see exactly where to cut by observing the reflection of the label pattern on the side of a scissors blade; if the cut is too far away from the edge of a label, a sliver of white will appear in the reflection.

It is advantageous to first cut the pattern into four strips of four labels each. Then the adhesive of choice can be applied to the back (unprinted) surface of the label strips. Then the scissors can be used to cut each strip into its four labels, being careful to NOT let the adhesive-backed labels fall onto the table, workbench, etc. In fact, each separated label, with its adhesive backing, may be carefully placed at an angle against some hard raised surface, so the adhesive has minimal contact with other objects or surfaces. Then use a tweezers to lift one label by one of its corners (maybe best to use the lower right corner if a right-handed person is operating the tweezers), and carefully place it on the appropriate keycap face, adjusting alignment before pressing down to set the adhesive.

Comments on some adhesives that have been tried with these labels:

- Scotch “Extra Strength” permanent “tape runner”; this product dispenses a tacky line of permanent adhesive approximately 8mm wide. It is necessary to apply two overlapping lines of the adhesive on each 1x4 strip of key labels, and to extend the applied adhesive beyond the edges of the label strip; do this with the label strip resting face-down on an expendable piece of paper, and then use a scissors to carefully cut away the paper along the edges of the label strip, cutting off the excess adhesive overlap in the process. Then proceed to cut the strip into its four labels. Clean any adhesive from the blades of the scissors before repeating the above procedure on the other strips of labels. This works very well, has been used on a number of previous projects having similar key labels, and does seem to be permanent.
- Scotch “Permanent Glue Stick” does NOT work with the non-porous 3D printed keycaps and printed keycap labels.
- CA (“Super”) glue; it should work well, but is too hard to apply accurately and it likely to wick to undesired places.
- Various ‘permanent’ varieties of spray-adhesives such as those made by 3M/Scotch and other manufacturers. However, overspray can very easily get onto the faces of the labels, ruining them. For best results, tape the printed 4x4 label array, as a single piece and with the 2mm-wide outer border, face-down onto a piece of expendable paper, with the tape (e.g. “Scotch” tape or masking tape) being applied all the way around the edges of the label, but without the tape extending as far as the actual labels (NOT beyond that 2mm-wide outer margin); this will function as a spray-mask. After spraying, the label array may be cut away from the paper and then cut apart as described above.
- Generally, liquid or gel type adhesives are likely to cause problems.
- Although untested, it seems that “rubber cement” might work well if applied to the back of the entire 4x4 label array before proceeding to cut out the individual labels.

Red Display Filter

The H8 Mini kit does not include any filter material for the red LED display, but legibility of that display will be poor WITHOUT a suitable red filter. There are a few inexpensive options for kit builders to supply their own filters:

- The best option is a commercially produced red filter intended for red LED displays. The main manufacturer of this filter material for many decades was PRD Products, which solely produced ‘Lexan’ filters of 0.02” thickness and in colors including red. At the time of this writing, PRD is no longer in business and (most) sellers such as Digikey no longer have stock. The ideal size of filter is the 3.6” wide version, which can be cut down to fit the H8 Mini’s 2-3/4” x 9/16” display area (or 3” wide if it is desired to have the filter extend over the two discrete LED’s for “POWER” (D4) and “MON” (D5). The PRD part number for this filter is 6203030, and apparently PRD’s remaining stock is now distributed by a number of Chinese “NOS” (New Old Stock) warehouses, and may appear on eBay from time to time. After cutting to size (a sharp scissors or paper shear can be used), the filter can be affixed to the front of the LED displays (LED1 ~ LED3) using a few tiny dots of CA (Super) glue.
- Red-tinted ‘Pexiglas’ is available in small sizes and quantities from eBay sellers, although the red color usually available is not ideal for the wavelength of red LEDs, and the filter thickness will be several times thicker than the purpose-made red filter material described above. Also, while the purpose-made filter has a surface that minimizes reflections and glare, the Plexiglas will be very smooth and reflective. Note that Plexiglas and Lexan are NOT the same kind of material; see sidebar below.
- Amazon sells various sizes and quantities of red “gel” filter film, intended to change the color of light for theatrical situations. A typical example sold by Amazon is called “Colored Lighting Gel Filters, 10 Pcs Correction Gel Light Filter Transparent Color Overlays Film Plastic Sheets 8.5 x 11 Inch, for Film, Video, Photo, Stage (Red)” and a few sheets of the material can be purchased for a few dollars. This works well enough as a filter for red LED’s, but is very thin and flimsy, and thus will be more of a bother to affix to the LED display such that it lays flat without curling or wrinkling. It is also very reflective, so glare might be an issue.

Plexiglas (plexiglass) is a popular brand of “Poly” (Polymethyl methacrylate or PMMA), a synthetic polymer in the form of a transparent thermoplastic. Some other widely used brand names are Acrylite, Lucite, and Perspex.

Lexan is one of the several brands of polycarbonate.